

Advanced Data Science Project Report

(Reproducible Notebook + Diagnostics)

Loan Default Risk: Pricing and Default Prediction

Student:

MAYA ROSERO Manuel David

PROFESSOR:

ABOUCAYA William

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Executive Summary

This report analyzes a Loan Default Risk dataset through two complementary tasks: (1) regression to explain and predict Interest Rate (pricing), and (2) binary classification to explain and predict Loan Status (default risk). We implement reproducible preprocessing pipelines, run baselines and interpretable models, compare against non-linear ML benchmarks, and perform key diagnostics (residual analysis, heteroskedasticity, multicollinearity, calibration, and threshold policies).

The dataset contains 67,463 training rows and 28,913 test rows with 35 columns. Default prevalence in the training set is 9.25% (imbalanced classification).

Main finding: model choice was not the main limitation. Interest Rate regression achieved R-squared close to 0, and default prediction under origination-safe features produced ranking performance close to the class prevalence, suggesting limited predictive signal with the available pre-outcome features.

1. Problem Framing

Loan pricing and credit risk depend on borrower characteristics, credit history, and loan attributes. This project studies two related problems:

- Pricing (Regression): explain/predict Interest Rate.
- Risk (Classification): explain/predict Loan Status (0 = non-default, 1 = default).

Research questions:

- 1) Which variables explain Interest Rate?
- 2) Which variables explain/predict Loan Status?
- 3) Do ML models improve performance over interpretable baselines?
- 4) What are the main deployment caveats (imbalance, leakage, stability)?

Evaluation plan:

- Regression: MAE, RMSE, R-squared on a validation split.
- Classification (imbalanced): ROC-AUC, PR-AUC (Average Precision), Brier score, and confusion matrices under selected thresholds.

2. Data Loading and Quality Checks

We validated consistency between train and test (same columns and order), checked for duplicated columns, quantified missingness, and verified that Loan Status is fully missing in test (submission scenario).

Table 1 — Dataset overview (train vs test)

dataset	n_rows	n_cols	n_numeric_cols	n_object_cols
train	67463	35	26	9
test	28913	35	26	9

Table 2 — Loan Status class distribution (train)

Loan Status	count	pct
0.0	61222.0	90.75
1.0	6241.0	9.25

Table 3 — Top missingness rates in training data (top 12)

feature	Unnamed: 0	missing_rate
0	ID	0.0
26	Collection Recovery Fee	0.0
20	Revolving Utilities	0.0
21	Total Accounts	0.0
22	Initial List Status	0.0
23	Total Received Interest	0.0
24	Total Received Late Fee	0.0
25	Recoveries	0.0
27	Collection 12 months Medical	0.0
18	Public Record	0.0
28	Application Type	0.0
29	Last week Pay	0.0

3. Exploratory Data Analysis (EDA)

EDA provides descriptive insights about target distributions and key predictors. We focus on: (i) Interest Rate distribution, (ii) Default rate by Grade, and (iii) Loan Amount scale and skewness. EDA is descriptive and does not imply causality.

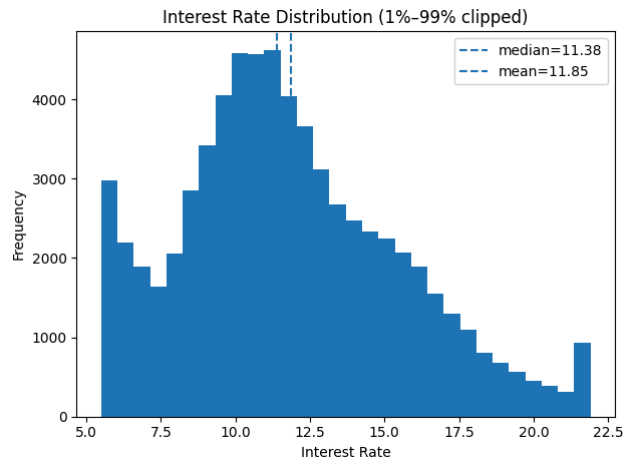


Figure 1 — Interest Rate distribution (1%-99% clipped).

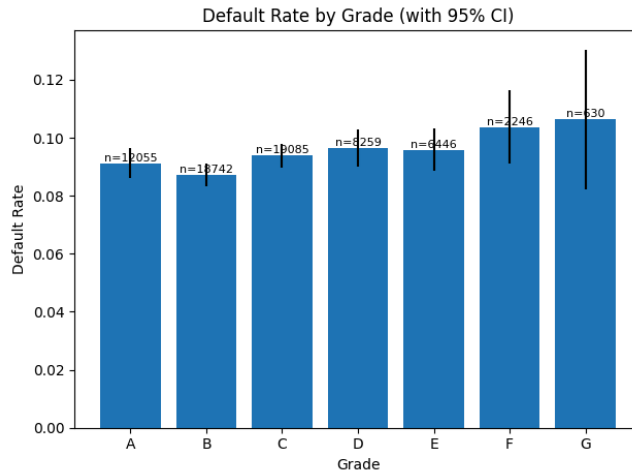


Figure 2 — Default rate by Grade (with 95% confidence intervals).

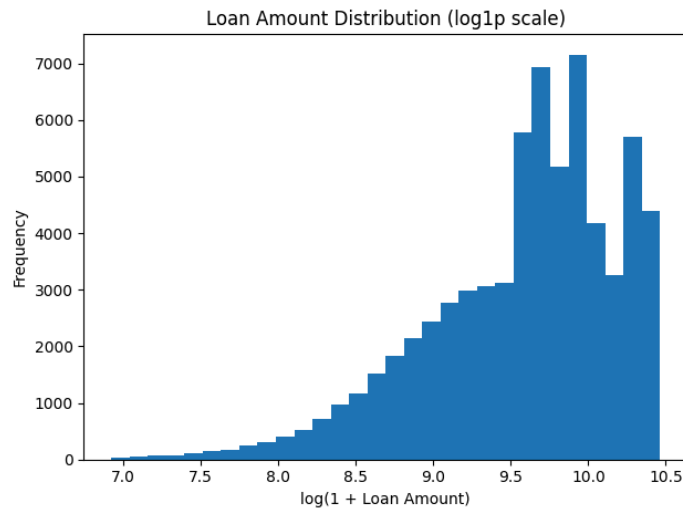


Figure 3 — Loan Amount distribution on $\log(1 + \text{amount})$ scale.

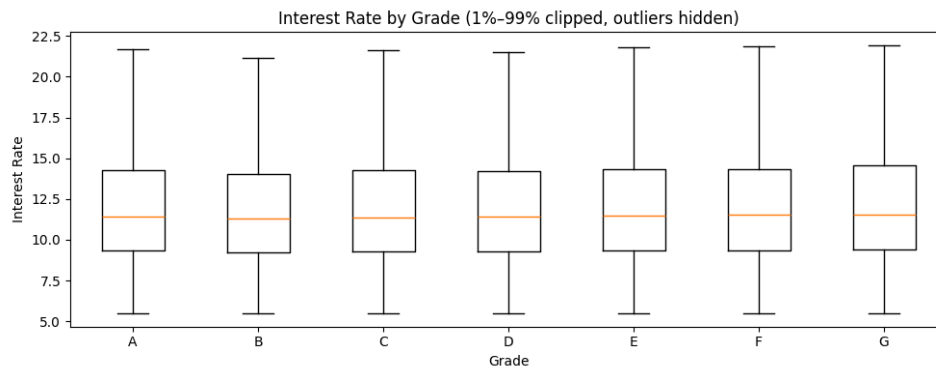


Figure 4 — Interest Rate by Grade (boxplot, clipped).

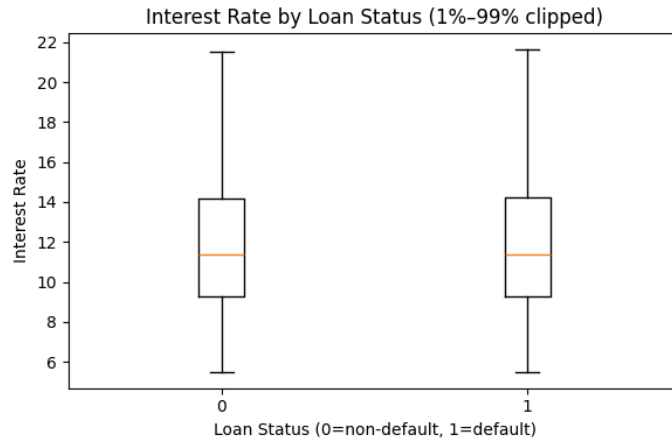


Figure 5 — Interest Rate by Loan Status (boxplot, clipped).

4. Preprocessing Strategy and Leakage Control

We implemented reusable scikit-learn preprocessing pipelines to ensure reproducibility: numeric imputation + scaling for linear models; one-hot encoding for nominal categories; and ordinal encoding for Grade/Sub Grade. For tree-based models we used a separate preprocessor that encodes all categorical features ordinally (no sparse matrices).

Leakage control (classification): we defined two feature sets:

- FULL: maximum information (may include outcome-adjacent variables).
- SAFE (origination-like): excludes leakage candidates detected via keyword scan (e.g., received, recoveries, collections) and excludes Interest Rate to avoid post-outcome contamination.

Table 4 — Feature schema summary by task and preprocessor

task	preprocessor	n_features	n_numeric	n_ordinal	n_categorical
regression	linear	30	22	2	6
cls_full	linear	31	23	2	6
cls_safe	linear	24	16	2	6
regression	tree	30	22	-	8
cls_full	tree	31	23	-	8
cls_safe	tree	24	16	-	8

Table 5 — Leakage candidate columns (keyword scan, top 15)

col	dtype	missing_rate
Collection 12 months Medical	int64	0.0
Collection Recovery Fee	float64	0.0
Recoveries	float64	0.0

Total Collection Amount	int64	0.0
Total Received Interest	float64	0.0
Total Received Late Fee	float64	0.0

5. Interest Rate Modeling (Regression)

We compared a naive mean baseline, regularized linear models (Ridge/Lasso), and a non-linear benchmark (HistGradientBoostingRegressor). Models were evaluated on a held-out validation split using MAE, RMSE, and R-squared.

Table 6 — Regression model comparison (validation)

model	MAE	RMSE	R2
HistGradientBoosting (tree preproc)	2.949574966487651	3.7086539789317294	0.0024056987965044
Lasso (best alpha=0.01)	2.9525105043359483	3.712421659605784	0.0003777252999812
Baseline (Mean)	2.953558148572899	3.7131817232918247	-3.163249229864995e-05
Ridge (best alpha=100.0)	2.953992560901197	3.7140405887147305	-0.0004943041236113

Best regression model by RMSE was: HistGradientBoosting (tree preproc) (MAE=2.950, RMSE=3.709, $R^2=0.0024$). However, R^2 remained close to 0, indicating limited explanatory power.

5.1 Regression diagnostics

We inspected standardized residuals vs fitted values and QQ-plots to assess misspecification and residual normality. We also ran the Breusch-Pagan test for heteroskedasticity and computed VIF on a reduced numeric/ordinal subset.

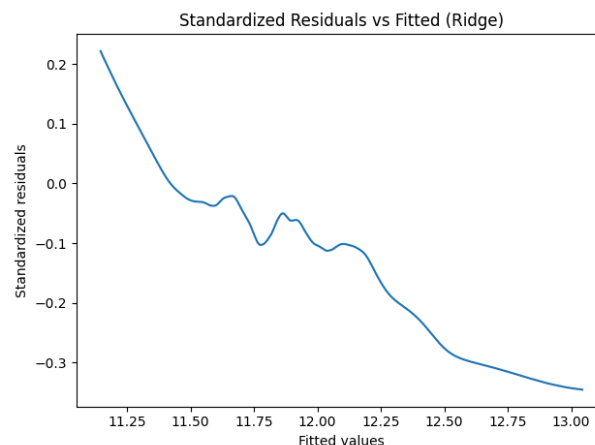


Figure 6 — Standardized residuals vs fitted (Ridge).

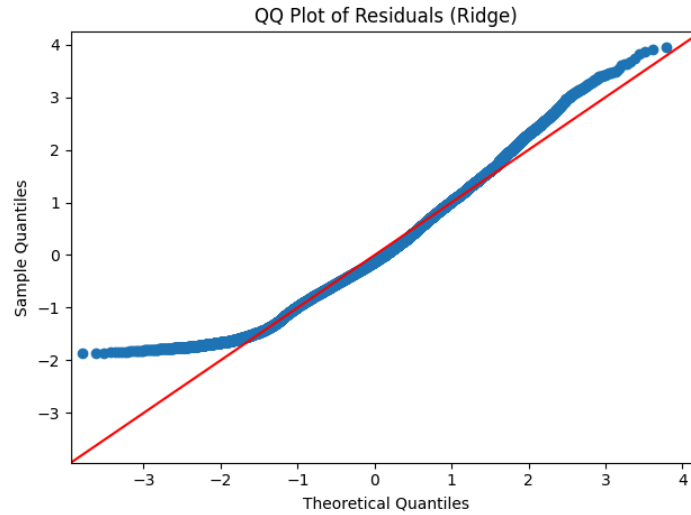


Figure 7 — QQ plot of residuals (Ridge).

Table 7 — Breusch-Pagan heteroskedasticity test (reduced OLS)

bp_stat	bp_pvalue	f_stat	f_pvalue	n_obs	n_features_reduced
94.84927141198204	2.2381128557771208e-10	3.956150579751784	2.197734611677853e-10	67463.0	24.0

Table 8 — VIF (top 12)

feature	VIF
Grade	1.0044481610648173
Revolving Balance	1.002976665960044
Total Accounts	1.0028295674817045
Total Revolving Credit Limit	1.0028184548538082
Home Ownership	1.002718038495392
Open Account	1.001692158410919
Public Record	1.001601860385375
Term	1.001599527404572
Debit to Income	1.0015645697650462
Revolving Utilities	1.001451927826627
Sub Grade	1.001307666428152
Total Current Balance	1.001281747845873

6. Loan Default Modeling (Classification)

Loan Status prediction is an imbalanced classification problem. Therefore, accuracy alone is misleading: a model predicting the majority class can achieve high accuracy while failing to detect defaults. We prioritize threshold-free ranking metrics (ROC-AUC, PR-AUC) and calibration (Brier score), and we evaluate operational thresholds using policies such as top-k risk selection and max-F1.

Table 9 — Classification ranking metrics (validation)

ROC_AUC	PR_AUC	Brier	Prevalence	Prob_Mean	model
0.519823207 5363047	0.098831162 5212687	0.246281280 6912777	0.092492403 4684651	0.494766425 960973	HGBClassifier SAFE (sample_weight)
0.512701416 5907592	0.097131551 545188	0.084105555 0188068	0.092492403 4684651	0.092378020 4935073	LogReg FULL (unweighted)
0.511761407 0630608	0.096812525 2289784	0.084099529 4384692	0.092492403 4684651	0.092390313 6939779	LogReg SAFE (unweighted)
0.512143169 3731613	0.096800226 4814322	0.248376491 915881	0.092492403 4684651	0.495253670 1857747	LogReg FULL (balanced)
0.510684894 9335678	0.096453357 0683387	0.248490851 0599802	0.092492403 4684651	0.495506712 2055281	LogReg SAFE (balanced)
0.5	0.092492403 4684651	0.083937559 2511735	0.092492403 4684651	0.092514359 8295349	Baseline (prior)
0.5	0.092492403 4684651	0.092492403 4684651	0.092492403 4684651	0.0	Baseline (most_frequent)

Best non-baseline classifier by PR-AUC was: HGBClassifier SAFE (sample_weight) (ROC-AUC=0.520, PR-AUC=0.099). PR-AUC is close to prevalence (0.092), which indicates weak separability with SAFE features.

6.1 ROC/PR curves and confusion matrices

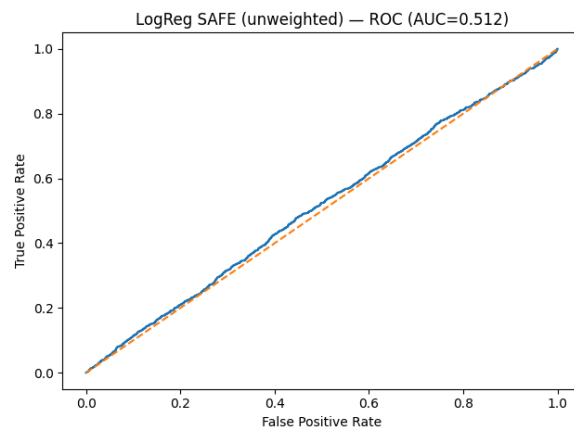


Figure 8 — ROC curve (LogReg SAFE, unweighted).

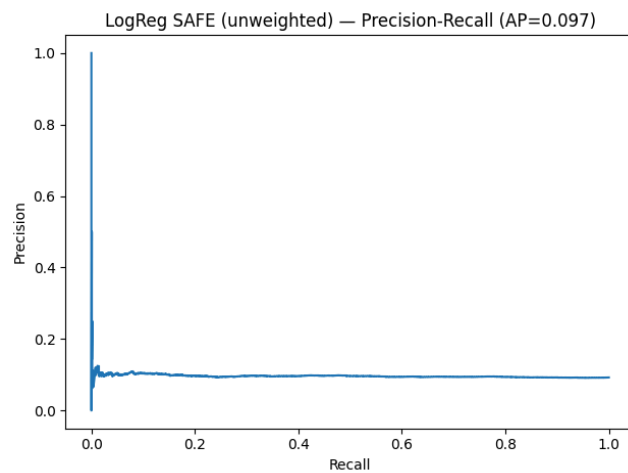


Figure 9 — Precision-Recall curve (LogReg SAFE, unweighted).

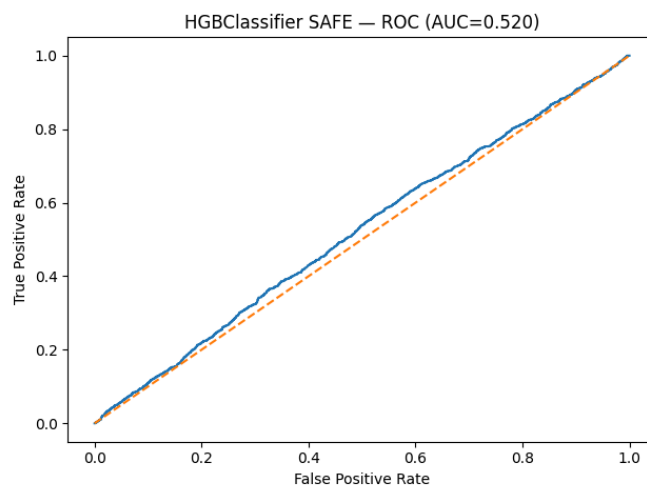


Figure 10 — ROC curve (HGBClassifier SAFE).

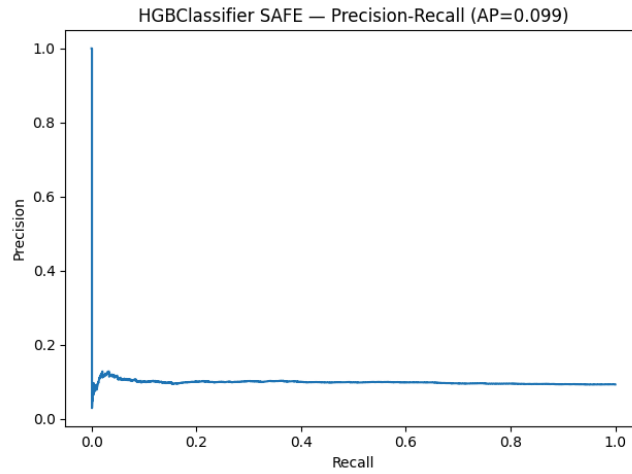


Figure 11 — Precision-Recall curve (HGBClassifier SAFE).

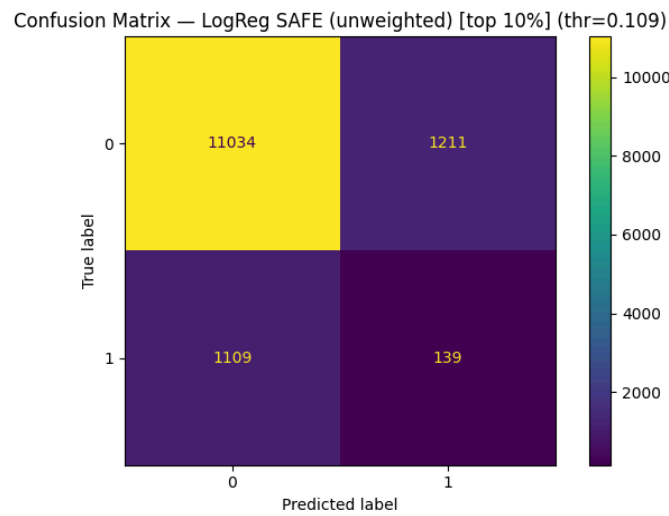


Figure 12 — Confusion matrix (LogReg SAFE, top-10% policy).

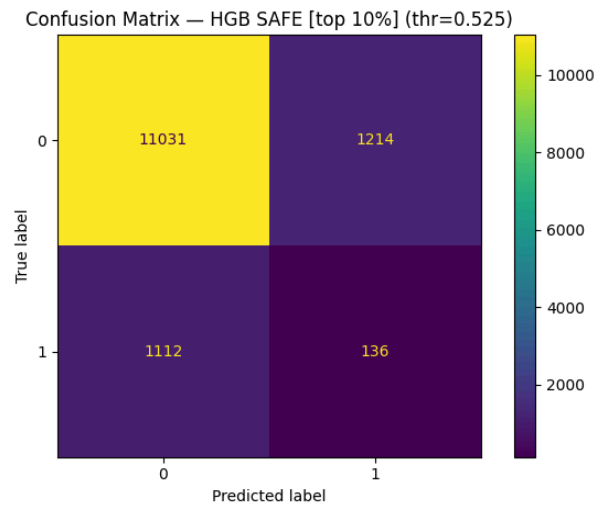


Figure 13 — Confusion matrix (HGBClassifier SAFE, top-10% policy).

6.2 Calibration

Calibration assesses whether predicted probabilities match observed default frequencies. We report Brier scores and show calibration curves. In our results, class weighting improves recall-related policies but may worsen probability calibration (higher Brier).

Table 10 — Brier scores (SAFE models)

model	Brier
LogReg SAFE (unweighted)	0.0840995294384692
LogReg SAFE (balanced)	0.2484908510599802

6.3 Feature importance (permutation)

Beyond coefficients, permutation importance provides a model-agnostic estimate of which raw features most impact Average Precision on the validation split (SAFE feature set).

Table 11 — Permutation importance (top 15 raw features, SAFE)

feature	importance_mean	importance_std
Loan Title	0.001145236786173	0.0005360715899015
Initial List Status	0.0011279593438897	0.0007080306984136
Grade	0.0010455243908432	0.0001204969849823
Batch Enrolled	0.00092312182818	0.0015043404397405
Employment Duration	0.0009069261887082	0.0005065420314137
Public Record	0.000777763813998	0.0006933792881503
Delinquency - two years	0.000693720714263	0.0003971922044605
Loan Amount	0.000293434391763	0.0001142828913594
Open Account	0.0001805277342629	0.0004055543702418
Revolving Utilities	0.0001276126457046	0.0001327363288522
Sub Grade	0.0001079202375755	0.0001062753112363
Total Current Balance	3.430954698480648e-05	0.0005080133441095
Application Type	3.2966405798959264e-05	3.592496741349383e-05
Inquires - six months	1.7449982884276682e-05	1.6280735540674236e-05
Funded Amount Investor	5.266572221385157e-07	9.927546330955736e-07

7. Model Improvement Attempts: High-Cardinality Handling and CatBoost

To improve performance, we explored CatBoost, which natively handles categorical variables and often performs well with high-cardinality categories and missingness. We also considered more robust one-hot settings (min_frequency and max_categories) to reduce sparsity and noise in linear models.

CatBoost achieved the strongest ROC/PR curves among the tested models in our validation split, but the improvement remained moderate, reinforcing that the limiting factor is likely the predictive signal available at origination time in this dataset.

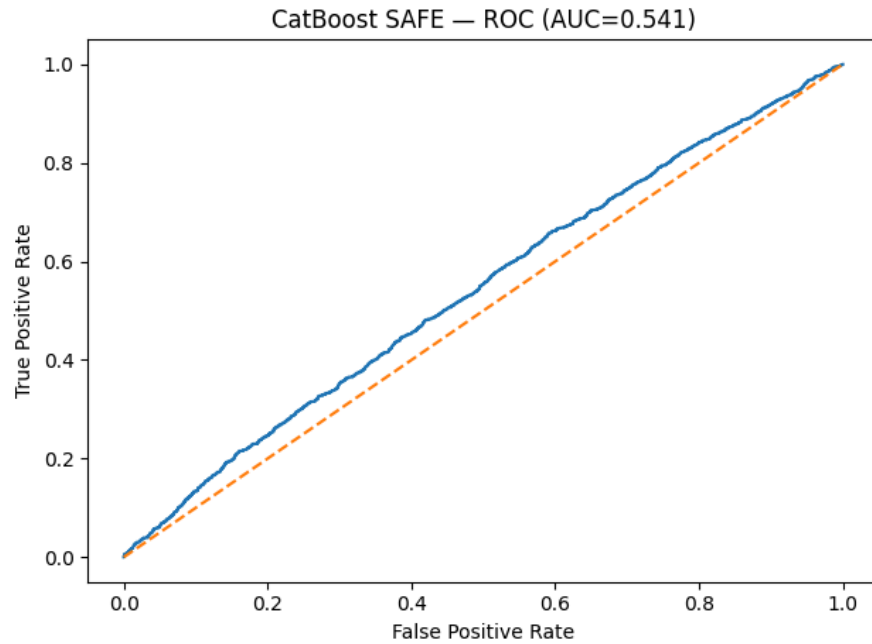


Figure 14 — ROC curve (CatBoost SAFE).

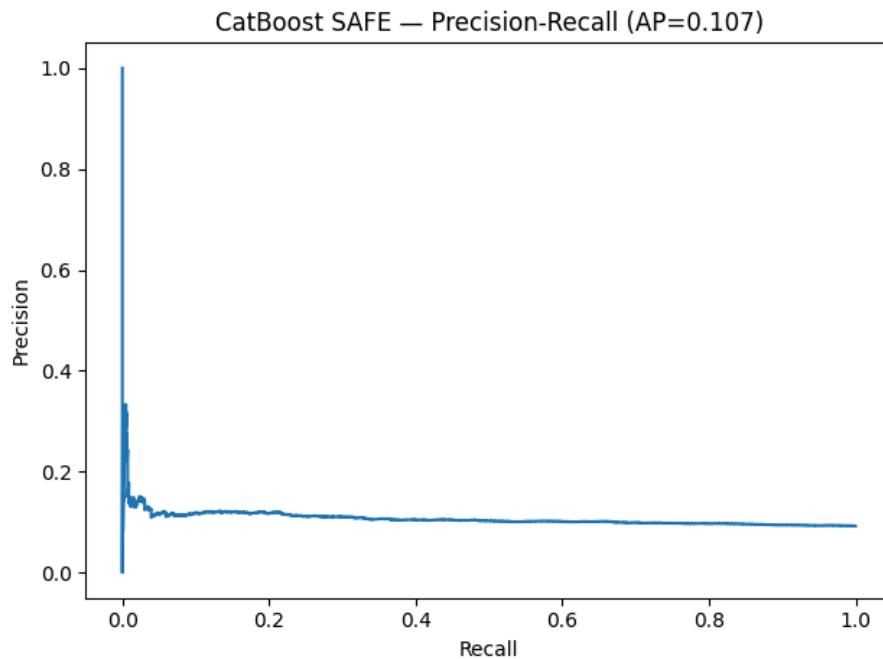


Figure 15 — Precision-Recall curve (CatBoost SAFE).

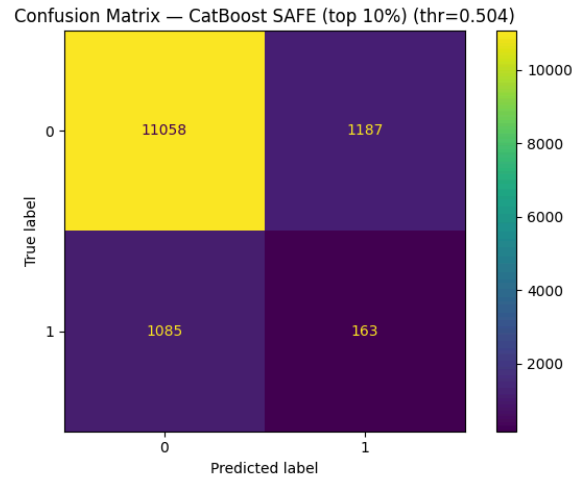


Figure 16 — Confusion matrix (CatBoost SAFE, top-10% policy).

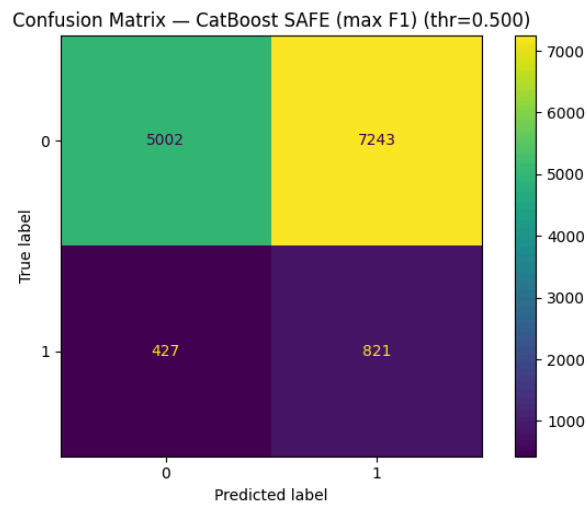


Figure 17 — Confusion matrix (CatBoost SAFE, max-F1 policy).

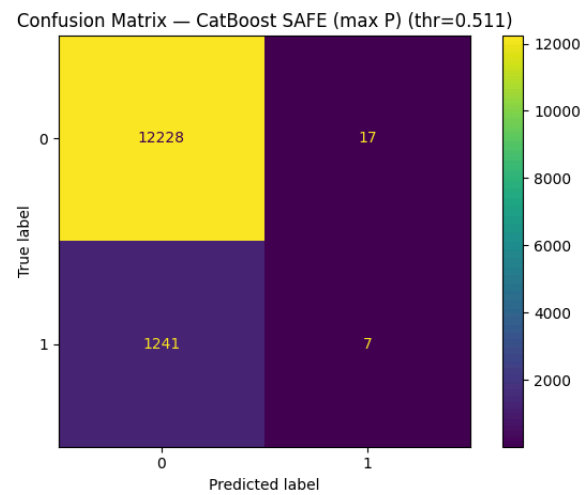


Figure 18 — Confusion matrix (CatBoost SAFE, precision-oriented policy)

8. Discussion: Why Predictive Performance Remained Low

Two aspects explain the perceived 'failure' to predict defaults with high accuracy:

- 1) Metric choice under imbalance: accuracy is threshold-dependent and can be misleading. When optimizing F1 or recall, the model predicts more positives, increasing false positives and reducing accuracy. A low accuracy (e.g., ~0.30) can therefore appear at aggressive thresholds even if ranking metrics are unchanged.
- 2) Weak separability: PR-AUC values close to the default prevalence indicate that the model's ranking is close to random. This suggests the available origination-safe features do not contain enough information to distinguish future defaulters from non-defaulters.

Additional likely contributors include: label noise or heterogeneous definitions of 'default', cohort effects (different time periods and underwriting rules), missing key behavioral variables, and limited interaction features. In such cases, more complex models cannot compensate for missing signal.

9. Conclusions and Next Steps

Regression (Interest Rate): All tested models achieved R-squared close to 0, indicating limited explanatory power for pricing with the available features. While models slightly improved MAE/RMSE relative to baseline, the overall signal remains weak.

Classification (Loan Status): Under leakage-controlled SAFE features, ranking performance was close to the baseline prevalence, implying weak predictability. CatBoost provided a modest improvement but did not change the overall conclusion.

Recommended next steps:

- Revisit the target definition (what constitutes default) and check for cohort/time leakage.
- Add richer features (payment history, utilization, behavior, time-on-book) if the goal is monitoring.
- If the goal is origination risk, focus on feature engineering, segment-specific models, and time-based validation.
- Report operational policies (top-k review, cost-sensitive thresholds) rather than accuracy.